

# CS101 Syllabus & Handbook

Department of Computer Science — Official syllabus.

Course title: Introduction to Algorithms (CS101).

Credits: 3. Prerequisites: CS100 (Intro Programming) and Discrete Math I.

# Learning Outcomes

- Analyze algorithms using asymptotic notation.
- Implement and compare fundamental sorting and searching algorithms.
- Apply BFS, DFS, and Dijkstra on suitable graphs.
- Formulate simple DP recurrences.

## Grading Breakdown

Homework: 25%. Midterm exam: 25%. Final exam: 35%. Participation/labs: 15%.

- Late homework: -10% per day, maximum 2 days.
- Lowest homework score is dropped.

## Exam Policies

Exams are closed-book except for one handwritten A4 cheat sheet.

- No calculators required; no networked devices.
- Academic integrity violations result in a zero and referral.

## Weekly Schedule (1–4)

Week 1: Asymptotics. Week 2: Searching. Week 3: Insertion & Merge sort. Week 4: Quicksort & heaps.

## Weekly Schedule (5–8)

Week 5: Hashing. Week 6: Graphs & BFS/DFS. Week 7: Shortest paths (Dijkstra). Week 8: Midterm.

## Weekly Schedule (9–12)

Week 9: DP intro. Week 10: Knapsack & LCS. Week 11: Amortized analysis. Week 12: Complexity survey & review.

## Required Textbook

CLRS — Cormen, Leiserson, Rivest, Stein: Introduction to Algorithms (3rd ed.), selected chapters.

Supplementary: course lecture slides and study notes distributed weekly.

## Office Hours

Instructor: Tue/Thu 14:00–15:00, Room 204.

TAs: Mon/Wed 16:00–17:30, Lab L3.

Forum response SLA: within 24 business hours.

## Lab Requirements

Six programming labs in Python or Java. Autograded with hidden tests.

- Lab 1: Big-O experiments. Lab 2: Binary search. Lab 3: Sorting.
- Lab 4: Hash map. Lab 5: BFS/DFS. Lab 6: DP knapsack.

## **Accessibility & Support**

Students needing accommodations should contact Disability Services by Week 2.

Mental health resources are listed on the department portal.

## Contact

Course email: [cs101-staff@university.example.edu](mailto:cs101-staff@university.example.edu)

Do not email homework solutions; use the official submission portal.